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huff: A Python package for Market Area Analysis

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Software

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Summary

Market area models, such as the *Huff Model* and its extensions, are used to estimate regional market shares and customer flows of retail and service locations. In health geography, market area and accessibility models are applied for the analysis of catchment areas and spatial accessibility of healthcare locations. The `huff` Python package provides a complete workflow for market area and spatial accessibility analysis, including data import, construction of origin-destination interaction matrices, basic model analysis, parameter estimation from empirical data, calculation of distance or travel time matrices, and map visualization. The package is modular and object-oriented. It is intended for researchers in economic geography, regional economics, marketing, geoinformation science, and health geography. The software is openly available via the Python Package Index (PyPI). Its development and version history are managed in a public GitHub Repository and archived at Zenodo. A methodological handbook is also available on GitHub Pages.

Statement of need

Research context

Market area models are used in economic geography, regional economics, geoinformation science, and marketing, enabling the analysis and forecasting of market areas and customer flows for retail and service locations. The classical and most popular approach is the *Huff Model* (Huff, 1962, 1963, 1964) and its numerous extensions, such as the *Multiplicative Competitive Interaction (MCI) Model* (Cooper & Nakanishi, 1983; Nakanishi & Cooper, 1974, 1982). Typical research applications include examining the influence of marketing or location variables on consumer store choice, forecasting the revenue of new locations, or predicting the impact of new locations on existing ones (De Beule et al., 2014; Fittkau, 2004; Li & Liu, 2012; Mensing, 2018; Oruc & Tihi, 2012; Suárez-Vega et al., 2015; Wieland, 2015, 2018).

In health geography, such models are used to analyse catchment areas with respect to medical practices and hospitals (Bai et al., 2023; Fülop et al., 2011; Jia, 2016; Latruwe et al., 2023; Rhein et al., 2025; Wieland, 2018). In both cases, methods for calculating spatial accessibility are used that are structurally related to market area models, such as the *Hansen Accessibility* (Hansen, 1959) and the *Two-Step Floating Catchment Area (2SFCA) Analysis* (W. Luo & Wang, 2003). The two concepts are also increasingly being linked to methods for analyzing the supply structure and accessibility of health locations (Liu et al., 2022; J. Luo, 2014; Rauch et al., 2023; Subal et al., 2021). These models are also applied to other location-related contexts such as airports or recreation facilities (H. Wang et al., 2022; Y. Wang et al., 2026).

Challenges in application

There are several major challenges in model-based market area and accessibility analyses:

- Model calibration based on observed data on consumer behavior and/or store sales is difficult because the models are nonlinear in their weighting parameters (Huff & McCallum, 2008; Wieland, 2017). For this purpose, the *MCI Model* (Cooper & Nakanishi, 1983;

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Nakanishi & Cooper, 1974, 1982) has been developed as an econometric estimation technique. As this approach requires empirical market shares for fitting, it is applied in cases where customer-store interaction data was obtained by surveys or secondary data (Baviera-Puig et al., 2016; Latruwe et al., 2023; Oruc & Tihi, 2012; Suárez-Vega et al., 2015; Wieland, 2015, 2018). When only total sales of the locations investigated are available, nonlinear iterative fitting approaches may be applied (De Beule et al., 2014; Güßefeldt, 2002; Haines Jr et al., 1972; Li & Liu, 2012; Liang et al., 2020; Mensing, 2018; Orpana & Lampinen, 2003; Wieland, 2017). Due to the pronounced sensitivity of market area models to weighting schemes, the availability of multiple calibration approaches is essential in market area analysis.

- Market area and accessibility models typically include travel times instead of Euclidean distances. Calculating travel times is based on graph theory network analysis and requires real street networks. Therefore, there is a need for GIS (Geographic Information System) support and/or access to an API providing calculations based on input origins and destinations (Huff & McCallum, 2008; Miller & Shaw, 2015). It is extremely helpful for researchers if they can also complete this part of the market area analysis workflow within the analysis tool.
- Researchers must choose and compare appropriate weighting functions and parameters, which may result in substantially different results. For input variables such as travel time, several weighting functions (e.g., power, exponential, logistic) are used, and the model results are compared using goodness-of-fit metrics (Bai et al., 2023; Latruwe et al., 2023; Li & Liu, 2012; Orpana & Lampinen, 2003). It is, thus, necessary that, within the market area analysis workflow, several weighting functions are available, and that researchers may compare different model specifications based on model evaluation metrics.

Functionality of the `huff` package

The `huff` package for Python v1.9.x essentially provides the following features:

- *Data management and preliminary analysis*: Users may load customer origins and supply locations from GeoDataFrames, point shapefiles, CSV, or XLSX files. Attributes of customer origins and supply locations (variables, weightings) may be set by the user. The next step is to create an *interaction matrix* with a built-in function, on the basis of which all implemented models can then be calculated. Within an interaction matrix, *transport costs* (distance or travel time between customer origins and supply locations) may be calculated with built-in methods.
- *Basic Huff Model analysis*: Given an interaction matrix, users may calculate probabilities and expected customer flows with respect to customer origins, and total market areas of supply locations.
- *Model fitting based on empirical data*: Given empirical data on customer flows, regional market shares, or total sales, users may calibrate market area models. Model parametrization may be conducted using the *MCI Model*, by Maximum Likelihood optimization, or by a local optimization algorithm [guessefeldt2002; wieland2017]. Additionally, machine learning-based market area models may be trained using empirical customer flow data. Evaluation metrics are available that are automatically calculated to assess the predictive fit of the models (e.g., *R-squared*, *RMSE*, *MAPE*).
- *Accessibility analysis*: The package provides methods of accessibility analysis, which may be combined with market area analysis (*Hansen Accessibility*, *2SFCA Analysis*). Competitor accessibility may also be calculated directly in order to extend the *Huff Model* in terms of the *Competing Destinations Model* (Fotheringham, 1985).
- *GIS tools*: The library includes auxiliary GIS functions for market area analysis (buffer, distance matrix, overlay statistics) and clients for OpenRouteService (Neis & Zipf, 2008)

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and OpenStreetMap (Haklay, 2008) for simple maps. All of them are implemented in the modeling functions but may also be used stand-alone.

State of the field

To the best of our knowledge, no open-source Python package currently provides market area analysis and parameter estimation for the Huff or MCI Model. In particular, there is no known open-source software package that covers the entire workflow of market area analyses, as described in the "Statement of need" section. Some but not all of the functionalities mentioned are implemented in R packages: Both the `SpatialPosition` package (Giraud & Commenges, 2025) and the `huff-tools` package (Pavlis et al., 2014) provide basic Huff Model analyses with two parameters, calculation of air distances, and map visualization. The R package `MCI` (Wieland, 2017) focuses on model fitting based on empirical data, but does not provide processing of geospatial data and the calculation of distances or travel times. Accessibility analysis via 2SFCA Analysis is implemented in the R package `accessibility` (Pereira & Herszenhut, 2024). The (almost) complete workflow for market area analyses using the Huff/MCI Model is currently only implemented in proprietary GIS software, namely the *ArcGIS Business Analyst* by *ESRI* (Esri, 2025; Huff & McCallum, 2008).

Software architecture and documentation

Design choices

The `huff` package is organized into a modular architecture that separates core modeling functionality from auxiliary helper modules. All model-related classes, methods and functions are implemented in the `models` module. Supporting functionalities are provided in separate modules, organized thematically. For example, the `ors` module provides an OpenRouteService client for retrieving travel time matrices and isochrones, which may be directly accessed from the `models` module. This design allows auxiliary functions to be used independently of the core models (stand-alone). Configurations for all models, classes, and functions are stored in a `config` module.

The `huff` library follows an object-oriented design. The class structure reflects the conceptual actors within a spatial market: Customer demand locations are represented by the `CustomerOrigins` class and supply locations by the `SupplyLocations` class. Their connection is established via an interaction matrix containing all possible origin-destination combinations and the corresponding data, such as travel times and location attributes. It is created from the location data using the built-in function `create_interaction_matrix()` from the `models` module, resulting in an instance of the `InteractionMatrix` class. All implemented model analyses can be performed from an `InteractionMatrix` object, with the individual steps of the model calculations being methods of this class, e.g., `transport_costs()` for adding distances or travel times, `probabilities()`, `flows()`, and `marketareas()` for Huff Model calculations, or `mci_fit()` for a MCI Model analysis. These model analyses return objects of specific classes for each model, e.g., `HuffModel` and `MCIModel` for Huff and MCI Models, respectively. This structure was chosen to ensure a consistent workflow and a unified data structure, regardless of which model analysis is to be performed. All mentioned classes include `summary()` and `plot()` methods. Any object contains time stamps for each change, which may be accessed with the `show_log()` method.

Example workflow

A basic Huff Model analysis (without empirical parameter estimation) in the `huff` package consists of the following steps: 1. Load geospatial data of customer origins and supply locations 2. Define their attributes and weightings 3. Create an interaction matrix from origins and destinations, including the calculation of distances or travel times 4. Calculate utilities, regional

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market shares, and expected customer flows 5. Calculate total market areas of all supply locations

Documentation

The workflow above is illustrated in the *Examples* section of the package README. For advanced model analyses (e.g., model calibration) and other package functionalities, the examples folder in the `huff` GitHub repository contains commented Python scripts. In addition to the API documentation and examples, `huff` provides a methodological handbook as a GitHub Pages site. This manual documents the mathematical formulations and theoretical foundations of the implemented models and provides guidance on their application and interpretation. This combination of software, examples, and methodological documentation is intended to support reproducible and transparent application of the package in research.

Research impact statement

The `huff` package has already been used in a health geography project at Würzburg University Hospital to model the catchment areas for pediatric oncology care; one publication citing the package has resulted from this research project so far (Kapitza et al., 2026). The library is actively used: since its first release in April 2025, it has been downloaded 44,915 times from the Python Package Index (source: pepy.tech, accessed October 5, 2026).

AI usage disclosure

No AI tools were used for software design, implementation, or decision-making. GitHub Copilot in Microsoft Visual Studio Code using the GPT-5 mini model (by OpenAI) was used to assist in drafting and refining docstrings for documentation. The corresponding guidelines and constraints defined by the author are documented in `AGENTS-docstrings.md` in the public GitHub repository. The manuscript text was written without the use of AI tools.

References

Bai, L., Tao, Z., Cheng, Y., Feng, L., & Wang, S. (2023). Delineating hierarchical obstetric hospital service areas using the Huff model based on medical records. *Applied Geography*, *153*, 102903. https://doi.org/10.1016/j.apgeog.2023.102903

Baviera-Puig, A., Buitrago-Vera, J., & Escriba-Perez, C. (2016). Geomarketing models in supermarket location strategies. *Journal of Business Economics and Management*, *17*(6), 1205–1221. https://doi.org/10.3846/16111699.2015.1113198

Cooper, L. G., & Nakanishi, M. (1983). Standardizing Variables in Multiplicative Choice Models. *Journal of Consumer Research*, *10*(1), 96–108. https://doi.org/10.1086/208948

De Beule, M., Van den Poel, D., & Van de Weghe, N. (2014). An extended huff-model for robustly benchmarking and predicting retail network performance. *Applied Geography*, *46*, 80–89. https://doi.org/10.1016/j.apgeog.2013.09.026

Esri. (2025). *ArcGIS business analyst*. https://www.esri.com/en-us/arcgis/products/arcgis-business-analyst/overview

Fittkau, D. (2004). *Beeinflussung regionaler Kaufkraftströme durch den Autobahnlückenschluß der A 49 Kassel-Gießen - Zur empirischen Relevanz der New Economic Geography in wirtschaftsgeographischen Fragestellungen* [Dissertation, Georg-August-Universität Göttingen]. https://doi.org/10.53846/goediss-3024

Fotheringham, A. S. (1985). Spatial Competition and Agglomeration in Urban Modelling. *Environment and Planning A: Economy and Space*, *17*, 213–230. https://doi.org/10.1068/a170213

DRAFT

Fülop, G., Kopetsch, P., & Schöpe, C. (2011). Catchment areas of medical practices and the role played by geographical distance in the patient's choice of doctor. *The Annals of Regional Science*, *46*(3), 691–706. https://doi.org/10.1007/s00168-009-0347-y

Giraud, T., & Commenges, H. (2025). *SpatialPosition: Spatial position models*. https://doi.org/10.32614/CRAN.package.SpatialPosition

Güßefeldt, J. (2002). Zur Modellierung von räumlichen Kaufkraftströmen in unvollkommenen Märkten. *ERDKUNDE*, *56*(4), 351–370. https://doi.org/10.3112/erdkunde.2002.04.02

Haines Jr, G. H., Simon, L. S., & Alexis, M. (1972). Maximum Likelihood Estimation of Central-City Food Trading Areas. *Journal of Marketing Research*, *9*, 154–159. https://doi.org/10.2307/3149948

Haklay, M. (2008). OpenStreetMap: User-generated street maps. *IEEE Pervasive Computing*, *7*(4), 12–18. https://doi.org/10.1109/MPRV.2008.80

Hansen, W. G. (1959). How Accessibility Shapes Land Use. *Journal of the American Institute of Planners*, *25*(2), 73–76. https://doi.org/10.1080/01944365908978307

Huff, D. L. (1962). *Determination of Intra-Urban Retail Trade Areas*. Real Estate Research Program, Graduate Schools of Business Administration, University of California.

Huff, D. L. (1963). A Probabilistic Analysis of Shopping Center Trade Areas. *Land Economics*, *39*(1), 81–90. https://doi.org/10.2307/3144521

Huff, D. L. (1964). Defining and estimating a trading area. *Land Economics*, *28*(4), 34–38. https://doi.org/10.2307/1249154

Huff, D. L., & McCallum, B. (2008). *Calibrating the Huff Model Using ArcGIS Business Analyst* (ESRI White Paper, September 2008). ESRI.

Jia, P. (2016). Developing a flow-based spatial algorithm to delineate hospital service areas. *Applied Geography*, *75*, 137–143. https://doi.org/10.1016/j.apgeog.2016.08.008

Kapitza, J., Wieland, T., & Metzler, M. (2026). Modeling hospital catchment areas in pediatric oncology using an empirically parameterized extended Huff model. *International Journal of Health Geographics*, *25*, 25. https://doi.org/10.1186/s12942-026-00478-2

Latruwe, T., Van der Wee, M., Vanleenhove, P., Michielsen, K., Verbrugge, S., & Colle, D. (2023). Improving inpatient and daycare admission estimates with gravity models. *Health Services and Outcomes Research Methodology*, *23*(4), 452–467. https://doi.org/10.1007/s10742-022-00298-4

Li, Y., & Liu, L. (2012). Assessing the impact of retail location on store performance: A comparison of wal-mart and kmart stores in cincinnati. *Applied Geography*, *32*(2), 591–600. https://doi.org/10.1016/j.apgeog.2011.07.006

Liang, Y., Gao, S., Cai, Y., Foutz, N. Z., & Wu, L. (2020). Calibrating the dynamic Huff model for business analysis using location big data. *Transactions in GIS*, *24*(3), 681–703. https://doi.org/10.1111/tgis.12624

Liu, L., Lyu, H., Zhao, Y., & Zhou, D. (2022). An Improved Two-Step Floating Catchment Area (2SFCA) Method for Measuring Spatial Accessibility to Elderly Care Facilities in Xi'an, China. *International Journal of Environmental Research and Public Health*, *19*, 11465. https://doi.org/10.3390/ijerph191811465

Luo, J. (2014). Integrating the Huff Model and Floating Catchment Area Methods to Analyze Spatial Access to Healthcare Services. *Transactions in GIS*, *18*(3), 436–448. https://doi.org/10.1111/tgis.12096

Luo, W., & Wang, F. (2003). Measures of Spatial Accessibility to Health Care in a GIS Environment: Synthesis and a Case Study in the Chicago Region. *Environment and*

DRAFT

Planning B: Planning and Design, *30*(6), 865–884. https://doi.org/10.1068/b29120

Mensing, M. (2018). *Lebensmittel-Onlinehandel - Alternative zur zukünftigen Versorgung der Bevölkerung ländlicher Räume?* (pp. 1 Online–Ressource (x, 333 Seiten) : Illustrationen, Diagramme, Karten) [Dissertation, Rheinisch-Westfälische Technische Hochschule Aachen]. https://doi.org/10.18154/RWTH-2019-02683

Miller, H. J., & Shaw, S.-L. (2015). Geographic Information Systems for Transportation in the 21st Century. *Geography Compass*, *9*, 180–198. https://doi.org/10.1111/gec3.12204

Nakanishi, M., & Cooper, L. G. (1974). Parameter estimation for a Multiplicative Competitive Interaction Model: Least squares approach. *Journal of Marketing Research*, *11*(3), 303–311. https://doi.org/10.2307/3151146

Nakanishi, M., & Cooper, L. G. (1982). Technical Note — Simplified Estimation Procedures for MCI Models. *Marketing Science*, *1*(3), 314–322. https://doi.org/10.1287/mksc.1.3.314

Neis, P., & Zipf, A. (2008). OpenRouteService.org is three times "open": Combining OpenSource, OpenLS and OpenStreetMap. *GIS Research UK Conference (GISRUK 2008)*.

Orpana, T., & Lampinen, J. (2003). Building spatial choice models from aggregate data. *Journal of Regional Science*, *43*(2), 319–348. https://doi.org/10.1111/1467-9787.00301

Oruc, N., & Tihi, B. (2012). Competitive Location Assessment - the MCI Approach. *South East European Journal of Economics and Business*, *7*(2), 35–49. https://doi.org/10.2478/v10033-012-0013-7

Pavlis, M., Dolega, L., & Singleton, A. (2014). *Huff-tools*. https://github.com/alexsingleton/Huff-Tools/

Pereira, R. H. M., & Herszenhut, D. (2024). *Accessibility: Transport accessibility measures*. https://doi.org/10.32614/CRAN.package.accessibility

Rauch, S., Stangl, S., Haas, T., Rauh, J., & Heuschmann, P. U. (2023). Spatial inequalities in preventive breast cancer care: A comparison of different accessibility approaches for prevention facilities in bavaria, germany. *Journal of Transport & Health*, *29*, 101567. https://doi.org/10.1016/j.jth.2023.101567

Rhein, M. von, Hauser, J., Haldimann, L., Jörg, R., & Gruebner, O. (2025). Imbalanced access to pediatric primary care in Switzerland: geographic differences and modeled future challenges. *European Journal of Pediatrics*, *184*, 648. https://doi.org/10.1007/s00431-025-06441-w

Suárez-Vega, R., Gutiérrez-Acuña, J. L., & Rodríguez-Díaz, M. (2015). Locating a supermarket using a locally calibrated Huff model. *International Journal of Geographical Information Science*, *29*(2), 217–233. https://doi.org/10.1080/13658816.2014.958154

Subal, J., Paal, P., & Krisp, J. M. (2021). Quantifying spatial accessibility of general practitioners by applying a modified huff three-step floating catchment area (MH3SFCA) method. *International Journal of Health Geographics*, *20*, 9. https://doi.org/10.1186/s12942-021-00263-3

Wang, H., Wei, X., & Ao, W. (2022). Assessing Park Accessibility Based on a Dynamic Huff Two-Step Floating Catchment Area Method and Map Service API. *ISPRS International Journal of Geo-Information*, *11*(7). https://doi.org/10.3390/ijgi11070394

Wang, Y., Lin, L., Meng, X., Zhu, M., & Kan, C. (2026). Measuring airport catchment areas via the Huff gravity model calibrated with mobile location data—Evidence from the Yangtze River Delta region. *Journal of Transport Geography*, *131*, 104552. https://doi.org/10.1016/j.jtrangeo.2026.104552

Wieland, T. (2015). *Räumliches Einkaufsverhalten und Standortpolitik im Einzelhandel unter Berücksichtigung von Agglomerationseffekten - Theoretische Erklärungsansätze,*

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modellanalytische Zugänge und eine empirisch-ökonometrische Marktgebietsanalyse anhand eines Fallbeispiels aus dem ländlichen Raum Ostwestfalens/Südniedersachsens. MetaGIS. ISBN: 978-3-936438-73-4

Wieland, T. (2017). Market area analysis for retail and service locations with MCI. *The R Journal*, *9*, 298–323. https://doi.org/10.32614/RJ-2017-020

Wieland, T. (2018). Modellgestützte Verfahren und "big (spatial) data" in der regionalen Versorgungsforschung II: Räumliche Interaktionsmodelle. *Monitor Versorgungsforschung*, *11*(3), 59–64. https://doi.org/10.24945/MVF.03.18.1866-0533.2083

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